

Team binomial / protocol 0.3.0-demo / CNY reference prices 2026-08, electronics ~108, OEM toy not included.

Ref	Qty	Part	Use / constraint	CNY
U1	1	ESP32-S3-N16R8 DevKit	Only MCU. Adv Nascent-Toy. Does not drive motors	48.00
U2	1	DHT11	GPIO4. Ambient T/H, not contact fuse	4.00
U3	1	MPU6050 / GY-521	I2C GPIO8/9 @0x68. Insert inference; motor sense RO	9.00
U4	1	FSR402	GPIO1 ADC. No climax detect. Right unpopulated	22.00
U5	1	WS2812B x8	GPIO6. 5V. Level = lit count	10.00
Q1	1	AO3400A N-MOS SOT-23	Parallel OEM key. 1=G 2=S 3=D	0.80
R1	1	1k 1/8W	GPIO7 series to gate	0.05
R2	1	47k 1/8W	Gate pulldown. Solder before flying leads	0.05
R3	1	10k 1/8W	Drain limit. Safety part, do not omit/reduce	0.05
R4	1	10k 1/8W	FSR pulldown	0.05
R5	1	4.7k 1/8W	DHT11 pullup; omit if module has it	0.05
W1	1	flying leads	OEM key contacts. Not on breadboard	6.00
J1	1	USB-C cable	Power and flash	8.00
SW1	1	BOOT GPIO0	On-board. Short=e-stop, hold 2s=clear latch	0
HOST	1	OEM toy	OEM MCU + button + 3 motors. Not in purchase total	-

Not populated this demo: NTC x2, FSR right, e-stop lanyard, I2S mic. HW504 / K10 deleted.

Pins: GPIO4 DHT11, GPIO1 FSR, GPIO8/9 MPU, GPIO6 LED, GPIO7 AO3400 gate, GPIO0 BOOT.

CSV: bom.csv Schematic: schematic.svg / schematic.png and sheet 2 of this PDF.

